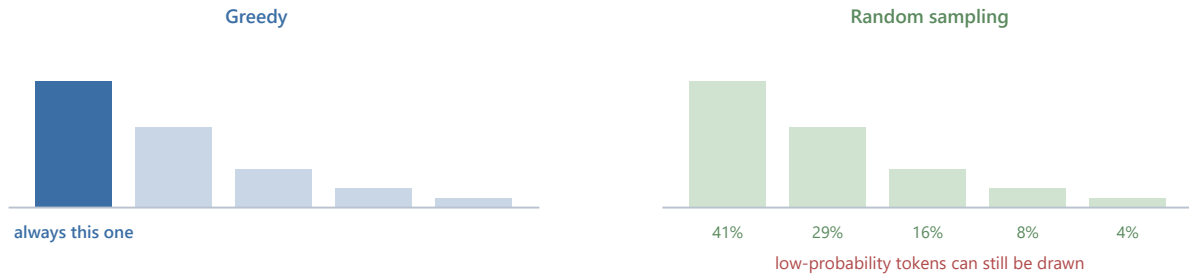


Random sampling

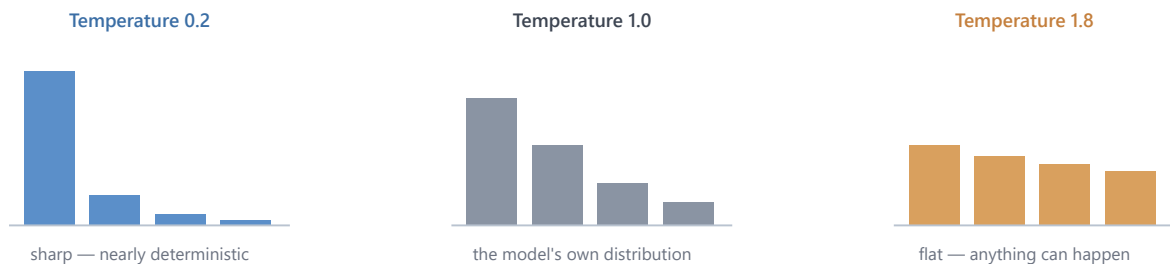
Draw from the distribution instead. A token with 41% is picked 41% of the time. This gives you variety — and also gives every one of those 50,000 near-zero tokens a small but nonzero chance of being picked. That is where a lot of nonsense comes from.



Greedy takes the tallest bar every time. Sampling rolls a weighted die.

Temperature

Temperature reshapes the distribution *before* sampling. Low temperature makes the tall bars taller and the short bars shorter — the model becomes more confident and more boring. High temperature flattens everything — more creative, more likely to say something strange. At temperature 0 you are back to greedy.



Temperature doesn't change what the model knows. It changes how strongly the model's confidence is expressed.

Top-K

Keep only the K most likely tokens and throw the rest away, then sample from those. This kills the long tail of nonsense. The problem is that K is a fixed number and confidence is not. If the model is sure, K=5 drags in four bad options. If the model is genuinely torn between twenty reasonable words, K=5 cuts off fifteen good ones.

Top-P (nucleus sampling)

Instead of counting tokens, count probability. Sort the tokens by likelihood and keep adding them to the pool until their combined probability crosses P (say 90%). Sample from that pool. When the model is confident, the pool is tiny. When it's unsure, the pool grows on its own. This is why top-P is usually the better default.